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DCNN: a novel binary and multi-class network intrusion detection model via deep convolutional neural network

Ahmed Shebl^{1*}, E. I. Elsedimy², A. Ismail¹, A. A. Salama¹ and Mostafa Herajy¹

Abstract

Network security become imperative in the context of our interconnected networks and everyday communications. Recently, many deep learning models have been proposed to tackle the problem of predicting intrusions and malicious activities in interconnected systems. However, they solely focus on binary classification and lack reporting on individual class performance in case of multi-class classification. Moreover, many of them are trained and tested using outdated datasets which eventually impact the overall performance. Therefore, there is a need for an efficient and accurate network intrusion detection system. In this paper, we propose a novel intelligent detection system based on convolutional neural network, namely DCNN. The proposed model can be utilized to efficiently analyze and detect attacks and intrusions in intelligent network systems (e.g., suspicious network traffic activities and policy violations). The DCNN model is applied against three benchmark datasets and compared with state-of-the-art models. Experimental results show that the proposed model improved resilience to intrusions and malicious activities for binary as well as multi-class classification, expanding its applicability across different intrusion detection scenarios. Furthermore, our DCNN model outperforms similar intrusion detection systems in terms of positive predicted value, true positive rate, *F1* measure, and accuracy. The scores obtained for binary and multi-class classifications on the CICIoT2023 dataset are 99.50% and 99.25%, respectively. Additionally, for the CICIDS-2017 dataset, DCNN attains a score of 99.96% for both binary and multi-class classifications, while the CICIoMT2024 dataset attains a score of 99.98% and 99.86% for binary and multi-class classifications, respectively.

Keywords Convolutional neural network, Cybersecurity, Deep learning, Deep neural network, Network intrusion detection

1 Introduction

The tremendous expansion in networks as well as the exponential development of technologies and services related to the Internet of Things (IoTs), cloud computing and big data has led to users' excessive reliance on these

services and technologies in their daily activities [1]. As a result, the confidentiality and integrity of sensitive data has become vulnerable to serious forms of intrusions, leaks and potential threat of unauthorized access by malicious individuals [2]. Moreover, the advancement in today's technologies and the hackers' determination to continuously enhance these penetrations and cyber-attacks have given rise to novel and unexpected intrusions which are increasingly sophisticated [3, 4]. Therefore, enhancing network security and defending it from such complex intrusions and hacking attempts has been a growing trend among network researchers in recent years [5–7]. This makes intrusion detection an

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important research field that requires in-depth investigation and has led to the conception of network intrusion detection systems (NIDSs) [8]. Through their ability to detect and predict intrusions, NIDS technology has proven its outstanding reliability in blocking hacking attempts by unauthorized users and sustaining network security [9, 10].

Correspondingly, artificial intelligence (AI) [11] that simply refers to machines imitating the intelligent human behavior [12], is increasingly becoming a key feature in the field of network security and intrusion detection [13–16]. Machine learning (ML) and deep learning (DL) algorithms being utilized for learning and development, enabling the examination of massive datasets and the detection of subtle insights and patterns that surpass human ability, thereby enhancing NIDSs [17, 18]. DL models are types of ML that takes inspiration from human's brain with its ability to filter information by utilizing stacked mathematical functions arranged in layers to mimic neurons' functions of the biological neural networks achieving depth and hence deep learning [19]. These models have the ability to efficiently learn the concealed network behavior and features, extract patterns and non-redundant information from data. Thus, this approach enables efficient detection of potential attacks in large datasets, unlike ML, which works better with smaller datasets [20, 21].

Nevertheless, despite the noticeable advancements in attack detection techniques for IoT environments, there are still many limitations in current proposed models. On the one hand, some existing research primarily emphasizes binary classification (e.g., see [7, 22, 23]). However, this approach is limited, as different attack types require individual analysis and investigation. On the other hand, other researchers may direct their attention towards multi-class classification (e.g., see [5, 24–31]), disregarding the notion of binary classification. Furthermore, monitoring the performance of each class is imperative in order to guarantee that the model is functioning effectively across all class. However, many models lack reporting on the performance of individual classes in multi-class classification (e.g., see [6, 26, 27, 29, 32–37]). Hence, resulting in an incomplete comprehension of the model's performance. Moreover, another issue of the currently proposed NIDS models is the utilization of outdated datasets during model training and testing which do not accurately reflect modern IoT network traffic (e.g., see [6, 22, 28, 29]). Thus, leading to poor model generalization and inaccurate model performance metrics when deployed in real-world scenarios. Besides, the improper selection and adjustment of techniques and key parameters related to reducing dimensionality and sampling leads to defects in the entire process, such as potential

bias, overfitting and the exclusion of important information. As a result, classification algorithms suffer degraded performance, particularly for minority samples.

In this paper, convolutional neural network (CNN) and deep neural network (DNN) models are utilized to propose an NIDS based on deep learning. This method combines the strengths of both deep learning architectures: CNNs excel at identifying important patterns by treating rows or columns as spatial features, while DNNs capture complex relationships between features, especially in high-dimensional data or when feature interactions are key. This results in a more accurate model with improved performance in both multi-class and binary classification tasks.

Furthermore, as our proposed model targets the detection of ongoing malicious intrusions, it is best trained using three distinct yet closely related datasets: CICIOT2023 [33], CICIOT2024 [34], and CICSIDS-2017 [38]. These datasets are chosen due to the imbalance in class distribution, popularity, and the inclusion of a variety of network traffic examples, including both normal and malicious actions. In addition, the performance of our model is compared to that of the DNN as well as related state-of-art models. Experimental findings demonstrate that the proposed model exhibits increased *F*-score, accuracy and detection rate, indicating its robustness performance against cyber attacks.

The rest of this paper is organized as follow: We start with summarizing related work outlining recent DL models used for NIDS. Afterwards, the proposed model is presented thoroughly in Sect. 3. The experimental environment configuration and result assessment as well as the analysis of our experimental findings are presented in Sect. 4. Section 5 discusses the implications and the limitation of presented model. Finally, the paper is closed with conclusions and an outline for the future work.

2 Related work

Motivated by the need of detecting intrusions and defending networks against attacks, researchers have put a lot of efforts into developing a variety of network intrusion detection systems (NIDS) [5, 6, 24–29, 32, 35, 36]. Table 1 provides a summary of the limitations and strengths of recently published models. In the following, we further elaborate on the strengths and limitations of these models.

Various NIDS approaches offer promising results, but they are only tested and validated via multi-class classification. For instance, recently, authors in [5] proposed a hybrid IDS based on DL by merging CNN and LSTM (long short-term memory network) while optimizing category weights to reduce data imbalance. Similarly, in [24], an IDS based on CNN is proposed to attain the

Table 1 Summary of the limitations and strengths (in table footnotes) of existing IDS models

#	Limitations	Model
1	Limited to binary classification	[7] ^{a,b} , [22] ^{a,b} and [23] ^f
2	Limited to multi-class classification	[5] ^f , [24] ^{a,d,f} , [25] ^{a,d,f} , [26] ^{a,f} , [27] ^{a,b,f} , [28] ^{a,f} , [29] ^{a,f} , [30] ^{a,e,f} , and [31] ^{a,d}
3	Lack detailed reporting on the performance of individual classes in multi-class classification	[6] ^{a,b} , [26] ^{a,f} , [27] ^{a,b,f} , [29] ^{a,f} , [32] ^{a,f} , [33] ^{a,c} , [34] ^{a,c} , [35] ^{a,d} , [36] ^{a,f} , and [37] ^{a,d,f}
4	Possible risk of overfitting and time consumption due to the replication of minority class events	[6] ^{a,b} , [7] ^{a,b} , [25] ^{a,d,f} , and [29] ^{a,f}
5	Important information may be excluded during features' reduction	[22] ^{a,f} and [35] ^{a,f}
6	Trained and tested using outdated datasets	[6] ^{a,b} , [22] ^{a,f} , [28] ^{a,f} , and [29] ^{a,f}

^a Offer a comparative analysis^b Provide novel feature selection^c Utilize novel datasets and benchmarks^d Offer real-time detection^e Exhibit explainability and transparency of the model outcomes^f Make use of hybrid models

security of network against intrusions, while in [25], an IDS based on Robust Transformer named RTIDS, is introduced. In addition, a two-stage hybrid model based on DL named LSTM-AE is proposed in [26], while in [30], an innovative approach was introduced. That approach focus on enhancing attack detection through leveraging a blending model, offering high accuracy and a deeper understanding of attack classifications. In addition, authors in [27] proposed a one-dimensional Pyramid Depth-wise Separable CNN IDS (PyDSC-IDS) based on OneHot Encoding and VGM, while another intrusion detection method is introduced in [29]. In a similar manner, authors in [31] introduced IoT-PRIDS, novel packet representation-based approach for intrusion detection. This approach is a lightweight non-ML model that can operate in near real-time scenarios, as it only requires parsing packets.

Furthermore, many of the presented models do not provide in-depth information regarding the performance of individual classes in the context of multi-class classification. Examples of models that belong to this category include those introduced in [26, 27, 29]. Likewise, authors in [6] proposed a hierarchical CNN-Attention network named CANET with Equalization Loss v2 in order to balance learning attention of minority classes by increasing their weights. This proposal was found to enhance the rate of detecting minority classes. In addition, a hybrid IDS combining the feature extraction ability of the LSTM and the spatial feature extraction ability of the CNN is proposed in [32], while in [35], a bi-phase NIDS that utilizes a combination of PSO and GA for feature selection is outlined. However,

there is a possibility to exclude important information when reducing features. Another novel technique for intrusion detection in IoT named DIS-IoT is proposed in [36]. The proposed model utilizes a stacking ensemble of four DL models integrated into a deep fully connected layer. In the same way, in [37] an innovative model, namely, CKAN (Convolutional Kolmogorov-Arnold Network) is proposed. The CKAN model is developed through replacing the MLPs (Multi-Layer Perceptrons) layers with KANs layers inside the CNN architecture.

Moreover, some approaches are limited to binary classification. For example, in [22], an approach to select optimal features is proposed by combining statistical importance through the difference in standard deviation of median and mean. Moreover, in [23], a two-phase IDS is introduced to classify multi-class data using Naive Bayes classifier, followed by majority voting. In the second phase, binary data are passed through an unsupervised elliptic envelope for classification. Another approach to intrusion detection based on AutoML and a soft voting is proposed in [7]. This approach utilizes an AutoML framework to select supervised classifiers and sampling methods to address data imbalance and minority classes.

Additionally, some developed models adopts sampling techniques to solve the problem of data imbalance. However, although sampling techniques serves as a solution to the problem of data imbalance, the replication of minority class events may increase the risk of overfitting and time consumption. As examples of models that fit into this category are those proposed in

[6, 7, 25, 29]. Notably, important information may be excluded during features' reduction as in [22, 35].

Furthermore, the majority of recent research models in the domain of NIDS including [6, 22, 28, 29] have employed outdated testing datasets, which may not accurately reflect the real-world data. Therefore these models may not be trained or tested on the most relevant and up-to-date data. This could affect the model's ability to learn new patterns or features, potentially impacting its accuracy and reliability when deployed. Therefore, in order to support the development of security analytical applications in actual IoT operations, a new and broad IoT attack dataset that comprises several attacks which are not present in existing IoT datasets is provided in [33]. Similarly, a novel and realistic dataset comprising wide range of attacks for Internet of Medical Things (IoMT) security is introduced in [34]. Several ML and DL models were used to examine and test the datasets in both studies. However, no information is provided regarding the performance of individual classes in the context of multi-class classification.

Nevertheless, despite advancements in attack detection techniques for IoT environments, Table 1 highlights that there are still exist persistent limitations in current network intrusions detection models. To address these issues, we propose in this paper a hybrid deep convolutional neural network (DCNN) model for classifying network attacks, which is thoroughly analyzed and evaluated for both binary-class and multi-class classification, with detailed reporting and analysis of individual class performance.

3 Methodology

This section outlines the fundamental design principles that form the basis of the proposed model. For the purpose of training our model, the CICIoMT2024, CICIDS-2017, and CICIoT2023 benchmark datasets are presented and analyzed. Afterwards, the structure of the datasets is provided, along with the preprocessing techniques employed. Finally, the details of the proposed deep neural network model is given.

3.1 Datasets

For the purpose of evaluating the performance of our model, three benchmark datasets were employed: CICIoMT2024¹ [34], CICIoT2023² [33], and CICIDS-2017³ [38]

3.1.1 CICIoT2023

The CICIoT2023 dataset represents a novel and extensive IoT attacks dataset which is publicly available. It consists of 33 attacks in total executed in an IoT topology composed of 105 devices. These attacks are classified into seven categories and contains about 47 features as outlined in Table 2.

3.1.2 CICIDS-2017

The CICIDS-2017 dataset composed of benign as well as a range of common intrusions, featuring at total of 14 class of attacks as outlined in Table 2. This representation of real-world scenarios is based on five days of Canadian Cybersecurity Institute (CIC) traffic data captured in eight files. The labeled network traffic, which is publicly available, contains over 80 features.

3.1.3 CICIoMT2024

The CICIoMT2024 dataset serves as a realistic benchmark for evaluating the security within the healthcare sector. This dataset is publicly available, containing about 46 features, while encompassing a diverse set of 18 distinct cyberattacks as illustrated in Table 2.

3.2 Data preprocessing

Figure 1 depicts the steps we employed in our data preprocessing phase. As the CSV files of the two datasets are publicly accessible, we can load and merge all files into a single data frame, facilitating efficient analysis and preprocessing in preparation for training.

To enhance the dependability and accuracy of the dataset, it is imperative to include a data cleaning phase within the data preprocessing pipeline, wherein errors, inconsistencies, and inaccuracies are identified and removed. Handling irrelevant, inconsistent and redundant data, dropping null and infinite values, locating and eliminating duplicates and checking for data integrity problems are all included in this stage. Additionally, the One-Hot Encoder was employed to convert non-numerical label values into numerical values for compatibility with deep neural networks, while unnecessary features like timestamp, flow id, destination ip, and source ip were excluded.

Features that substantially vary in scale do not have equal impact on analysis, potentially causing bias. Given that the features within the input dataset exhibit wide variations within their ranges, and due to the fact that the dataset is characterized by a high level of imbalance, the data standardization approach is adopted to normalize the input dataset by centering values around the mean (μ) with a unit standard deviation (σ) using (1).

¹ <https://www.unb.ca/cic/datasets/iomt-dataset-2024.html>

² <https://www.unb.ca/cic/datasets/iotdataset-2023.html>

³ <https://www.unb.ca/cic/datasets/ids-2017.html>

Table 2 Summary of the CICIoT2023, CICIDS-2017 and CICIoMT2024 datasets [33, 34, 38]

Class	CICIoT2023		Class	CICIDS-2017		Class	CICIoMT2024	
	Samples	%		Samples	%		Samples	%
DDoS	8,587,268	72.79	BENIGN	2,273,097	80.3004	TCP_IP-DDoS-UDP	1,635,956	22.85
DoS	2,043,697	17.32	DoS Hulk	231,073	8.1630	TCP_IP-DDoS-ICMP	1,537,476	21.47
Mirai	666,553	5.65	PortScan	158,930	5.6144	TCP_IP-DDoS-TCP	804,465	11.23
BenignTraffic	277,682	2.35	DDoS	128,027	4.5227	TCP_IP-DDoS-SYN	801,962	11.20
Spoofing	122,866	1.04	DoS GoldenEye	10,293	0.3636	TCP_IP-DoS-UDP	566,950	7.92
Recon	89,344	0.76	FTP-Patator	7938	0.2804	TCP_IP-DoS-SYN	441,903	6.17
Web - based	6278	0.05	SSH-Patator	5897	0.2083	TCP_IP-DoS-ICMP	416,292	5.81
Brute - Force	3318	0.03	DoS slowloris	5796	0.2048	TCP_IP-DoS-TCP	380,384	5.31
			DoS Slowhttptest	5499	0.1943	Benign	192,732	2.69
			Bot	1966	0.0695	MQTT-DDoS-Connect_Flood	173,036	2.42
			Web Attack - Brute Force	1507	0.0532	Recon-Port_Scan	83,981	1.17
			Web Attack - XSS	652	0.0230	MQTT-DoS-Publish_Flood	44,376	0.62
			Infiltration	36	0.0013	MQTT-DDoS-Publish_Flood	27,623	0.39
			Web Attack - Sql Injection	21	0.0007	Recon-OS_Scan	16,832	0.24
			Heartbleed	11	0.0004	ARP_Spoofing	16,047	0.22
						MQTT-DoS-Connect_Flood	12,773	0.18
						MQTT-Malformed_Data	5130	0.07
						Recon-VulScan	2173	0.03
						Recon-Ping_Sweep	740	0.01
Attacks	11,519,324	97.65	Attacks	557,646	19.6996	Attacks	6,968,099	97.31
Total	11,797,006	100	Total	2,830,743	100	Total	7,160,831	100

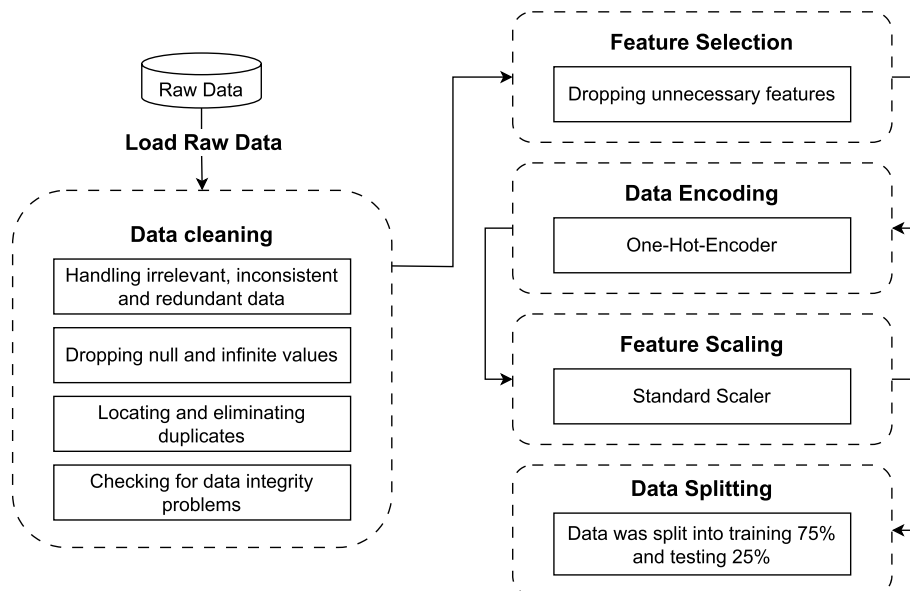


Fig. 1 A schematic diagram depicting the process of data preprocessing. First, we load and merge raw data. Then, errors, inconsistencies and duplicates are eliminated from the data in the cleaning phase. Unnecessary features are excluded during the feature selection phase. After that, non-numerical values are transformed into numerical values with the use of one-hot-encoder during the data encoding phase. Moreover, data is normalized in the feature scaling stage using standardization scaling technique. Finally, data was split into training 75% and testing 25%

$$X' = \frac{X - \mu}{\sigma} \quad (1)$$

3.3 Proposed model

The proposed hybrid model integrates the advantages inherent in CNNs and DNNs to enhance the performance of our network intrusion detection system (NIDS). The structure of this hybrid model is provided in Fig. 2, which encompasses 11 layers: an input layer, convolutional layers, activation layers, pooling layers, up-sampling layers, followed by fully connected deep layers and an output layer.

The first convolutional layer extracts features from the input data. First, the input sequence data undergoes convolutional layer processing, which serves as the fundamental element of the CNN. During the forward pass, the convolutional layer applies filters, known as kernels, in a sliding manner along the sequence. Meaning each of the N kernels will slide across the input sequence with a stride of S calculating the dot product between the sequence and the set of learnable parameters, or kernels at each position, resulting in the generation of a feature

map sequence representing the extracted features from the input sequence data.

This architectural design of CNNs [39] renders it possible to effectively handle and analyze data that exhibits a spatial or grid-like structure, such as image processing, natural language processing and even models involving sequential inputs. The CNN within the hybrid model focuses on extracting both sequential patterns and temporal features from the input traffic data. The architecture of the CNN component is composed of three 1D convolutional layers interspersed with max pooling and up-sampling layers. The convolutional layers employ a kernel size of three with a default stride of one, utilizing 256, 128, and 256 filters in each respective layer. The max-pooling layer operates with a pool size of three and a default stride of one, whereas the up-sampling layer utilizes an up-sampling factor of three.

Afterwards, an activation function is employed to process the feature map sequence by utilizing a non-linear activation function: the rectified linear unit (ReLU) [40], as given by (2).

$$Relu(x) = \max(0, x) \quad (2)$$

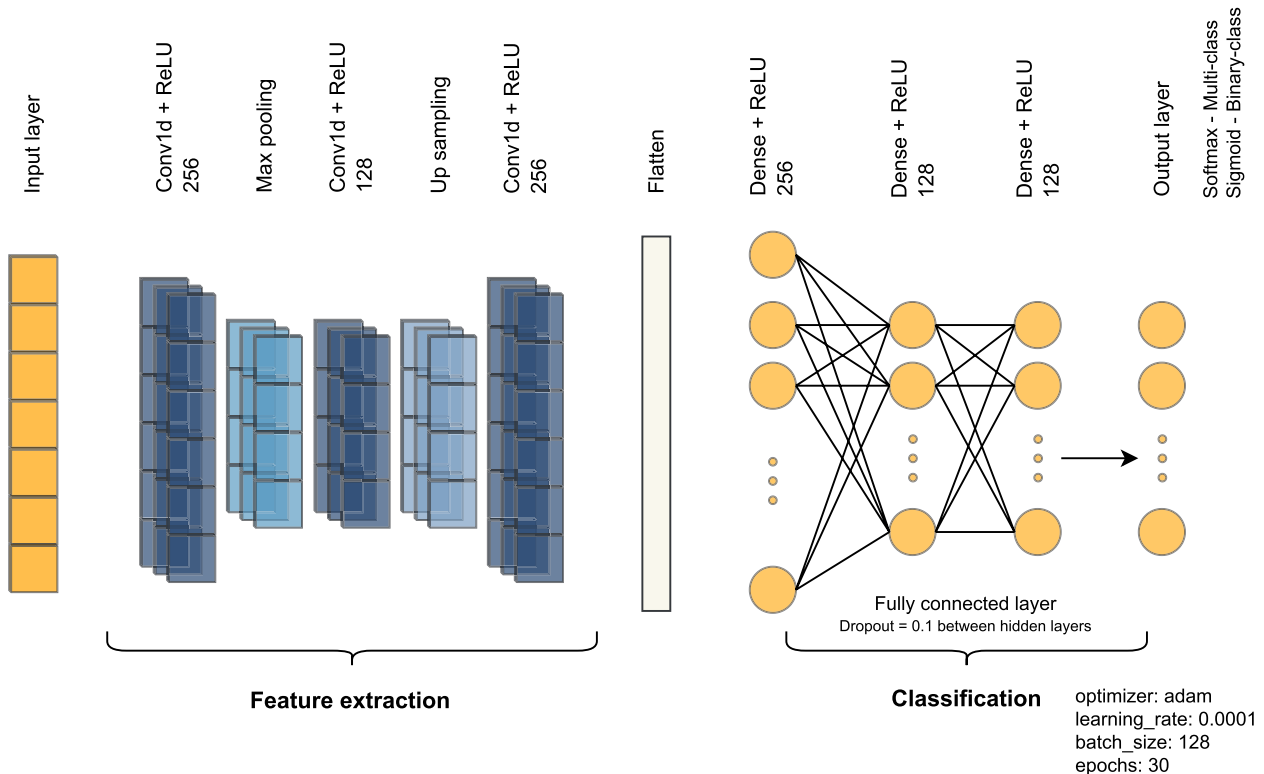


Fig. 2 The architecture of DCNN model merges the strengths of CNN and DNN models. CNN is exceptional in spatial and sequential feature extraction. The fully connected DNN layer acts as a powerful classifier. This integration enables a smooth and efficient workflow by using spatial features for precise categorization and labeling of input data

Pooling layers are utilized to compress the size of the generated map, thereby reducing parameters and computations while preserving significant features. Thus, exerting regulation over the challenge of overfitting. The max pooling process is applied to each window of size three along the sequence returning the maximum value. Conversely, up-sampling layers, which are essentially the antithesis of pooling layers, augment the dimensions of the map in preparation for the subsequent convolutional layer. The second and third convolutional layers extract higher-level features from the input sequence data, encompassing patterns and textures.

The DNN of the hybrid model is designed to classify the output of the CNN component into different classes of network traffic. The structural design of the DNN comprises of three fully connected layers (FCLs) each with 256, 128, and 128 neurons, respectively. Upon the completion of feature extraction for the training samples, the data undergoes flattening process and is subsequently fed into a sequence of deep FCLs. Each FCL employs the ReLU activation function, which is followed by a dropout layer that removes certain nodes from the network during training, thus eliminating their influence on prediction and back-propagation to mitigate the impact of overfitting, ultimately facilitating the classification process.

Subsequently, the output layer estimates the probabilities associated with each possible class as the final output. The output layer is equipped with the softmax function [12] defined by (3) in the context of multi-class classification, whereas in the scenario of binary-class classification, the sigmoid function [41] given by (4) is employed.

$$\text{Softmax}(x_i) = \frac{e^{x_i}}{\sum_{j=1}^K e^{x_j}} \text{ for } i = 1, 2, \dots, K \quad (3)$$

$$\text{Sigmoid}(x) = \frac{1}{1 + e^{-x}} \quad (4)$$

The CNN and DNN are integrated to create our hybrid model by flattening the output of the CNN part, then feeding it into the DNN part, which generates the final output of the model as shown in Fig. 2.

The model undergoes training through the Adam optimization algorithm with a learning rate set to 0.0001 (see Sect. 5 for more details). The binary crossentropy loss function is employed for binary-class classification tasks, whereas the categorical crossentropy loss function is utilized in the context of multi-class classification scenarios. Loss functions assess the extent to which the model's predictions diverge from the actual outcomes. A closer alignment of the model's predictions to the true values results in a minimal loss, while a significant deviation

from the original values yields a maximal loss. The training process comprises batch size of 128, with a total of 30 epochs being executed.

3.4 Evaluation metric

The performance assessment of the suggested NIDS model is accomplished by utilizing confusion matrix indicators. The following metrics are calculated as follows:

$$DR = TPR = \text{Sensitivity} = \text{Recall} = \frac{TP}{TP + FN} \quad (5)$$

$$PPV = \text{Precision} = \frac{TP}{TP + FP} \quad (6)$$

$$TNR = \text{Specificity} = \frac{TN}{TN + FP} \quad (7)$$

$$F1 - \text{score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (8)$$

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + FN + TN} \quad (9)$$

The acronyms TP, FP, FN, and TN represent true positive, false positive, false negative, and true negative. Moreover, DR and TPR in Eq. (5) represent detection rate and true positive rate, respectively, are metrics used to assess the performance of a classification model in the positive class. They quantify how well the model identifies positive instances correctly. PPV in Eq. (6) represents positive predictive value; it is a metric that focuses on the positive class and quantifies the number of actual positive samples out of all the samples predicted as positive. TNR in Eq. (7) is an abbreviation for true negative rate, it measures the model's ability to correctly identify negative instances. Moreover, Eq. (8) shows *F1*-score that is the harmonic mean of the PPV and TPR. Accuracy metric in Eq. (9), represents the total of correctly identified samples in our test data. In summary, these acronyms and metrics are crucial in evaluating the effectiveness and reliability of classification models and frequently employed when evaluating machine learning models.

4 Results

This section presents an assessment of the proposed model, encompassing a range of performance metrics such as accuracy, precision, recall, and *F1*-score (see Sect. 3.4). Furthermore, it showcases the visualization of the model's accuracy, model's loss, and confusion matrix, effectively illustrating its performance. In addition to this, it furnishes the outcomes of our testing, providing valuable insights into the identification and explication

of diverse attack types and normal behavior in two distinct scenarios: multi-class classification and binary-class classification. The primary objective is to comprehensively investigate the implications and outcomes associated with each approach and their overall impact on the analysis.

4.1 Experiment configuration

Table 3 outlines the details of the experimental design and setup that were employed in order to evaluate the various parameters of the model with utmost accuracy and *F*-score, facilitating the attainment of reliable and dependable results that can be used for further analysis and interpretation.

4.2 Experimental analysis

Table 4 provides a comprehensive overview of the performance exhibited by the proposed model using multi-class classification based on the CICIoT2023 dataset. It is worth noting that the proposed model demonstrates a good overall performance, achieving high accuracy rates consistently across various categories, with accuracy ranging from 99.47 to 100%. Moreover, upon closer examination of the results presented in Table 4, it

becomes apparent that certain classes within the dataset are exceptionally well-classified, particularly for those that possess a substantial number of instances within the dataset. The benign class demonstrates particularly impressive metrics, including a TPR of 95.02% and a TNR of 99.58%, indicating robust capabilities in identifying benign instances.

Furthermore, the model excels in detecting certain attacks with minuscule misclassification rate, notably DDoS, DoS, and Mirai, which achieves a remarkable overall metrics of over 99%, alongside a low FPR and FNR. Therefore, the proposed model exhibits the highest values in terms of DR and *F*-score for these classes. This highlights the effectiveness of the proposed model in accurately detecting and classifying instances pertaining to these specific attack types.

Furthermore, to gain further insights into the performance of the proposed model, it is crucial to examine the results depicted in the confusion matrix, as illustrated in Fig. 3c and d. The confusion matrix provides a visual representation of the performance of the model, showcasing the classification results for each class. Analyzing the confusion matrix enables a more nuanced understanding of the model's performance and the specific areas where improvements can be made. In order to provide a comprehensive overview of the findings, the performance results have been visually depicted in Fig. 3a and b. This visual representation serves as a succinct summary of the performance.

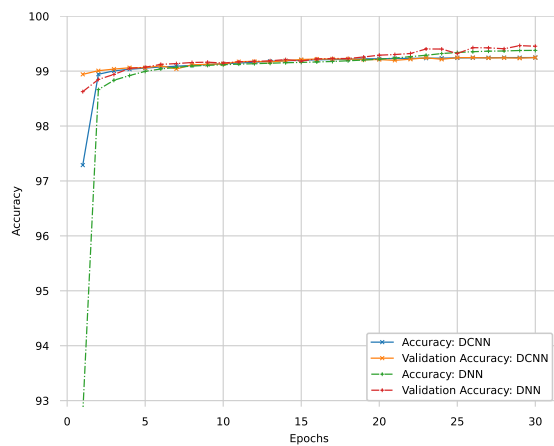
In contrast, for the case of binary-class classification using the CICIoT2023 dataset, one can observe a discernible improvement in the overall performance of the proposed model with respect to overall evaluation metrics, see Table 5. Both classes achieve good performance in terms of high TPR, accuracy, PPV, *F*-score, and TNR alongside a low FPR and FNR. The enhancements can be witnessed in a steady manner. The comprehensive

Table 3 Experiment configuration

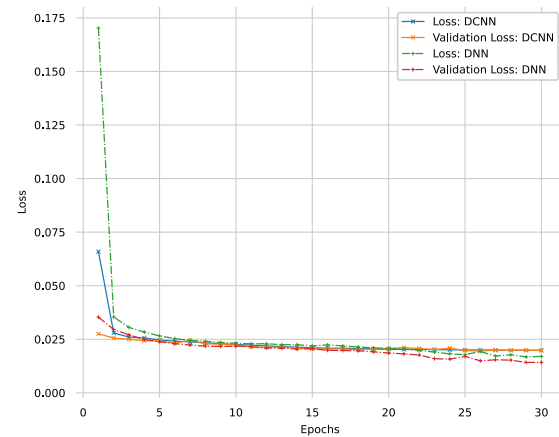
OS	Windows 11 Home
CPU	Intel Core i7 12700H 14core cash 24MB
GPU	NVIDIA RTX 3060 (6GB)
RAM	DDR5 16GB 4800 Mhz
Anaconda	23.7.4
Python	3.8.20
Tensorflow	2.7
Tensorflow-gpu	2.10

Table 4 Multi-class classification performance analysis with CICIoT2023

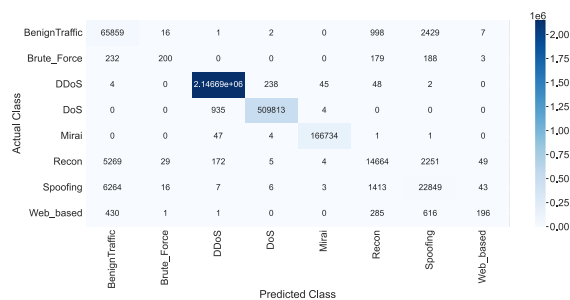
Class	Model	TPR	TNR	FPR	FNR	PPV	F1 measure	Accuracy
BenignTraffic	DNN DCNN	97.46% 95.02%	99.71% 99.58%	0.29% 0.42%	2.54% 4.98%	88.86% 84.37%	92.96% 89.38%	99.65% 99.47%
Brute - Force	DNN DCNN	19.83% 24.94%	100.00% 100.00%	0.00% 0.00%	80.17% 75.06%	96.36% 76.34%	32.89% 37.59%	99.98% 99.98%
DDoS	DNN DCNN	99.99% 99.98%	99.84% 99.86%	0.16% 0.14%	0.01% 0.02%	99.94% 99.95%	99.96% 99.97%	99.95% 99.95%
DoS	DNN DCNN	99.78% 99.82%	99.99% 99.99%	0.01% 0.01%	0.22% 0.18%	99.97% 99.95%	99.88% 99.88%	99.96% 99.96%
Mirai	DNN DCNN	99.98% 99.97%	100.00% 100.00%	0.00% 0.00%	0.02% 0.03%	99.98% 99.97%	99.98% 99.97%	100.00% 100.00%
Recon	DNN DCNN	78.70% 65.34%	99.89% 99.90%	0.11% 0.10%	21.30% 34.66%	84.50% 83.38%	81.50% 73.26%	99.73% 99.64%
Spoofing	DNN DCNN	78.70% 74.67%	99.90% 99.81%	0.10% 0.19%	21.30% 25.33%	89.08% 80.64%	83.57% 77.54%	99.68% 99.55%
Web - based	DNN DCNN	33.75% 12.82%	100.00% 100.00%	0.00% 0.00%	66.25% 87.18%	97.91% 65.77%	50.19% 21.46%	99.97% 99.95%



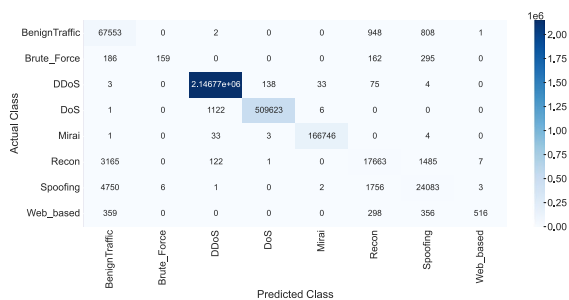
(a) The model accuracy curve illustrating the efficacy of the model in accurately predicting outcomes for both the training and validation data.



(b) Model loss curve representing the error and distance between the true values and the model's predicted values on both the training and validation data.



(c) Multi-class confusion matrix depicting distribution of the predictive performance of the DCNN



(d) Multi-class confusion matrix depicting distribution of the predictive performance of the DNN

Fig. 3 CICIOT2023 performance analysis in terms of multi-class classification (8 classes). The analysis includes visualizing accuracy and loss curves, alongside the confusion matrix of the proposed model, to assess the model's learning and improvement over time

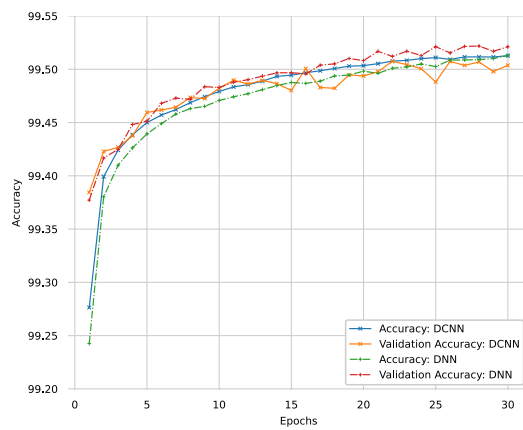
Table 5 Binary-class classification performance analysis with CICIOT2023

Class	Model	TPR	TNR	FPR	FNR	PPV	F1 measure	Accuracy
BenignTraffic	DNN DCNN	99.71% 99.68%	91.62% 92.16%	8.38% 7.84%	0.29% 0.32%	99.80% 99.81%	99.75% 99.75%	99.52% 99.50%
Attack	DNN DCNN	91.62% 92.16%	99.71% 99.68%	0.29% 0.32%	8.38% 7.84%	88.42% 87.41%	89.99% 89.72%	99.52% 99.50%

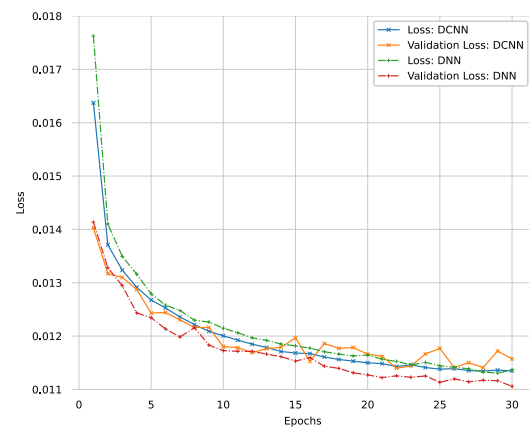
outcomes of the proposed model are visually represented in Fig. 4a depicting the model accuracy, and Fig. 4b showcasing the model loss providing a graphical representation of the model's performance. Additionally, the confusion matrix of the proposed model can be found in Fig. 4c and d.

Furthermore the classification process applied to the CICIDS-2017 dataset, which is detailed in Table 6, yielded highly favorable outcomes across multiple classes from the model proposed. The obtained results

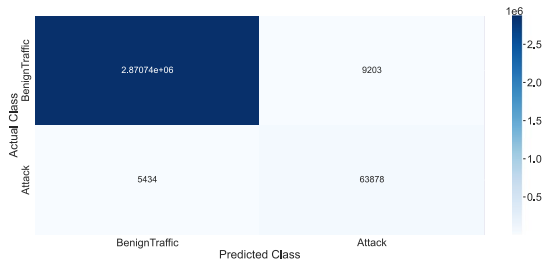
demonstrate that the suggested methods exhibit good performance in the scenario of multi-class classification with TPR, TNR and accuracy consistently exceeding threshold of 99% alongside very low FPR and FNR for the majority of classes. Notably, Heartbleed and Infiltration achieves the highest value of 100% in all metrics alongside a FPR and FNR of 0%, reflecting the model's robustness in detecting these specific minor classes. However, while these metrics showcase the model's strengths, a comprehensive examination



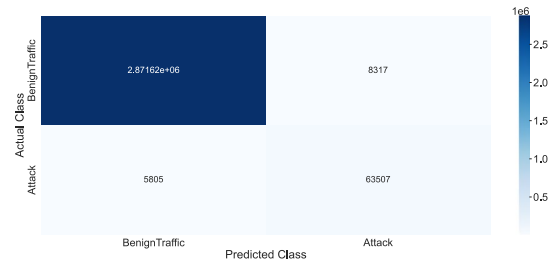
(a) The model accuracy curve illustrating the efficacy of the model in accurately predicting outcomes for both the training and validation data.



(b) Model loss curve representing the error and distance between the true values and the model's predicted values on both the training and validation data.



(c) Binary-class confusion matrix depicting distribution of the predictive performance of DCNN



(d) Binary-class confusion matrix depicting distribution of the predictive performance of DNN

Fig. 4 CICIOT2023 performance analysis in terms of binary-class classification (2 classes). The analysis includes visualizing accuracy and loss curves, alongside the confusion matrix of the proposed model, to assess the model's learning and improvement over time

reveals some notable limitations. For instance, the model's performance on Web Attack-Brute Force and Web Attack-XSS, where the TPR drops to 97.58% and 97.25%, respectively, in addition to PPV with 85.23% and 83.10%, and the FNR increases to 2.42% and 2.75%, respectively. Moreover, Web Attack-Sql Injection achieves a noticeable low TPR and F -score of 42.86% and 50% alongside a high FNR of 57.14%. This discrepancy can be attributed to the fact that the testing data comprises a relatively small number of instances pertaining to this specific attack type (also see Sect. 5).

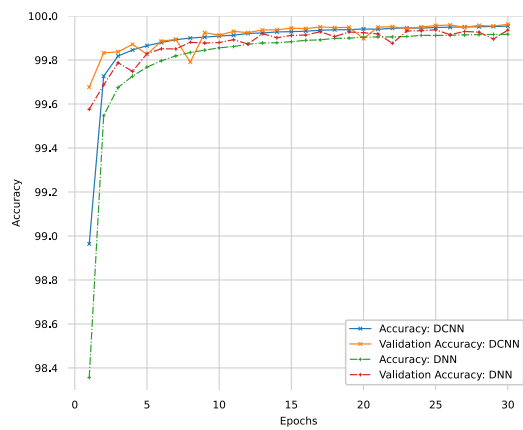
Nevertheless, it is important to highlight the robustness of the overall performance of the model. In order to provide a more comprehensive understanding of the model's performance, the resulting confusion matrix is presented in Fig. 5c and d. Additionally, Fig. 5a and b visually depict the model's performance in terms of both model accuracy and model loss. These visual representations serve to

enhance the clarity and comprehensibility of the obtained results, allowing for a more thorough evaluation of the proposed model's efficacy.

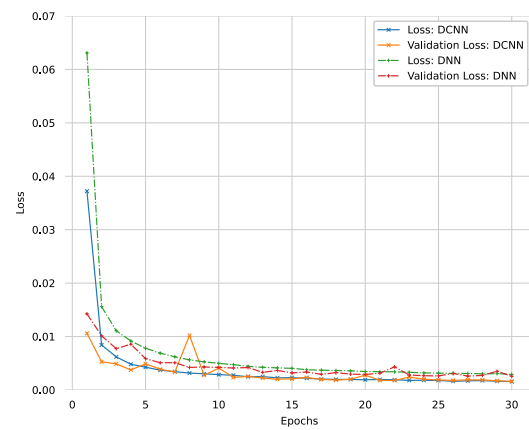
Furthermore, a similar trend emerges when binary-class classification is applied to the CICIDS2017 dataset, as showcased in Table 7. Once again, the proposed model exhibits a high level of performance in this context, further validating their effectiveness. The corresponding confusion matrix can be found in Fig. 6c and d, while Fig. 6a and b provide insights into the model's performance in terms of both model accuracy and model loss.

Moreover, the classification methodology of our model is also applied on the CICIOMT2024 dataset, as detailed in Table 8, yielded exceptionally favorable results across various categories derived from the proposed model. The acquired findings indicate that the recommended methodologies demonstrate good performance within the context of multi-class classification, with accuracy levels surpassing

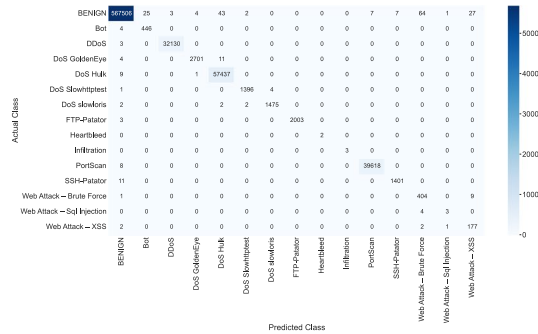
Class	Model	TPR	TNR	FPR	FNR	PPV	F1 measure	Accuracy
BENIGN	DNN DCNN	99.95% 99.97%	99.93% 99.97%	0.07% 0.03%	0.05% 0.03%	99.98% 99.99%	99.96% 99.98%	99.94% 99.97%
	DNN DCNN	96.00% 99.11%	100.00% 100.00%	0.00% 0.00%	4.00% 0.89%	98.86% 94.69%	97.41% 96.85%	100.00% 100.00%
DDoS	DNN DCNN	99.98% 99.99%	100.00% 100.00%	0.00% 0.00%	0.02% 0.01%	99.93% 99.99%	99.95% 99.99%	100.00% 100.00%
DoS GoldenEye	DNN DCNN	99.23% 99.45%	100.00% 100.00%	0.00% 0.00%	0.77% 0.55%	99.04% 99.82%	99.14% 99.63%	99.99% 100.00%
DoS Hulk	DNN DCNN	99.99% 99.98%	99.98% 99.99%	0.02% 0.01%	0.01% 0.02%	99.74% 99.90%	99.87% 99.94%	99.98% 99.99%
DoS Slowhttptest	DNN DCNN	99.00% 99.64%	100.00% 100.00%	0.00% 0.00%	1.00% 0.36%	98.72% 99.71%	98.86% 99.68%	100.00% 100.00%
DoS slowloris	DNN DCNN	98.99% 99.59%	100.00% 100.00%	0.00% 0.00%	1.01% 0.41%	99.19% 99.73%	99.09% 99.66%	100.00% 100.00%
FTP-Patator	DNN DCNN	99.65% 99.85%	100.00% 100.00%	0.00% 0.00%	0.35% 0.15%	99.75% 100.00%	99.70% 99.93%	100.00% 100.00%
Heartbleed	DNN DCNN	100.00% 100.00%	100.00% 100.00%	0.00% 0.00%	0.00% 0.00%	100.00% 100.00%	100.00% 100.00%	100.00% 100.00%
Infiltration	DNN DCNN	100.00% 100.00%	100.00% 100.00%	0.00% 0.00%	0.00% 0.00%	100.00% 100.00%	100.00% 100.00%	100.00% 100.00%
PortScan	DNN DCNN	99.93% 99.98%	100.00% 100.00%	0.00% 0.00%	0.07% 0.02%	99.99% 99.98%	99.96% 99.98%	100.00% 100.00%
SSH-Patator	DNN DCNN	98.87% 99.22%	100.00% 100.00%	0.00% 0.00%	1.13% 0.78%	99.22% 99.50%	99.04% 99.36%	100.00% 100.00%
Web Attack - Brute Force	DNN DCNN	99.28% 97.58%	99.99% 99.99%	0.01% 0.01%	0.72% 2.42%	85.63% 85.23%	91.95% 90.99%	99.99% 99.99%
Web Attack - Sql injection	DNN DCNN	28.57% 42.86%	100.00% 100.00%	0.00% 0.00%	71.43% 57.14%	100.00% 60.00%	44.44% 50.00%	100.00% 100.00%
Web Attack - XSS	DNN DCNN	95.05% 97.25%	100.00% 99.99%	0.00% 0.01%	4.95% 2.75%	85.64% 83.10%	90.10% 89.62%	99.99% 99.99%



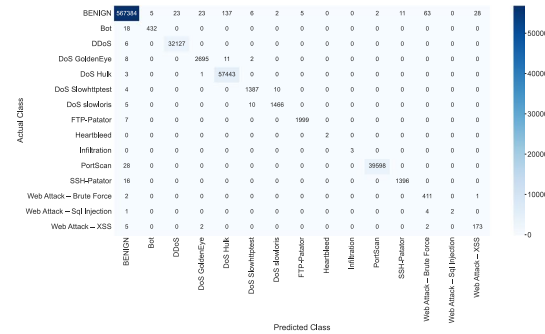
(a) The model accuracy curve illustrating the efficacy of the model in accurately predicting outcomes for both the training and validation data.



(b) Model loss curve representing the error and distance between the true values and the model's predicted values on both the training and validation data.



(c) Multi-class confusion matrix depicting distribution of the predictive performance of DCNN



(d) Multi-class confusion matrix depicting distribution of the predictive performance of DNN

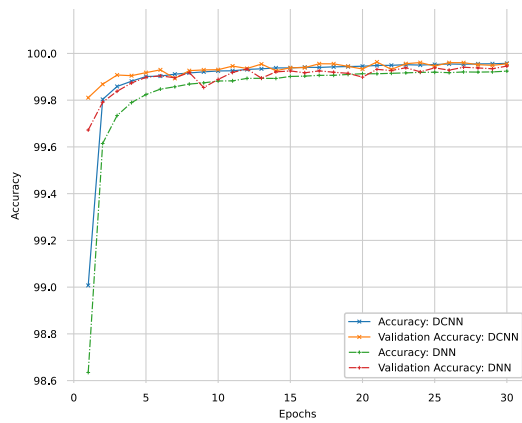
Fig. 5 CICIDS-2017 performance analysis in terms of multi-class classification (8 classes). The analysis includes visualizing accuracy and loss curves, alongside the confusion matrix of the proposed model, to assess the model's learning and improvement over time

Table 7 Binary-class classification performance analysis with CICIDS-2017

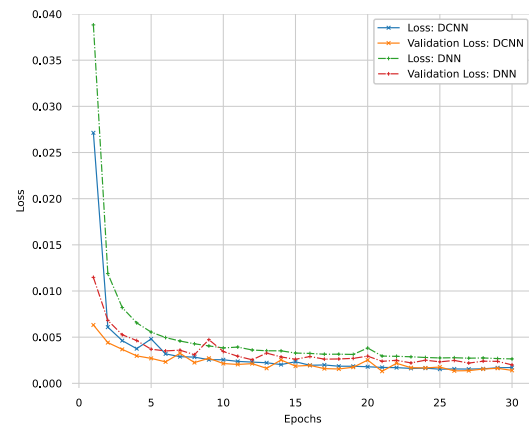
Class	Model	TPR	TNR	FPR	FNR	PPV	F1 measure	Accuracy
BENIGN	DNN DCNN	99.93% 99.94%	99.95% 99.96%	0.05% 0.04%	0.07% 0.06%	99.79% 99.83%	99.86% 99.89%	99.95% 99.96%
Attack	DNN DCNN	99.95% 99.96%	99.93% 99.94%	0.07% 0.06%	0.05% 0.04%	99.98% 99.99%	99.97% 99.97%	99.95% 99.96%

the threshold of 99.9%, and TPR and TNR predominantly exceeding the threshold of 99%, alongside remarkably FPR and FNR for the majority of classifications. This is particularly evident in the classes MQTT-DDoS-Connect_Flood, MQTT-DDoS-Publish_Flood, MQTT-DoS-Connect_Flood, DoS-Publish_Flood, TCP_IP-DDoS-ICMP, TCP_IP-DDoS-SYN, TCP_IP-DDoS-TCP, TCP_IP-DDoS-UDP, TCP_IP-DoS-ICMP, TCP_IP-DoS-SYN, TCP_IP-DoS-TCP, and TCP_IP-DoS-UDP.

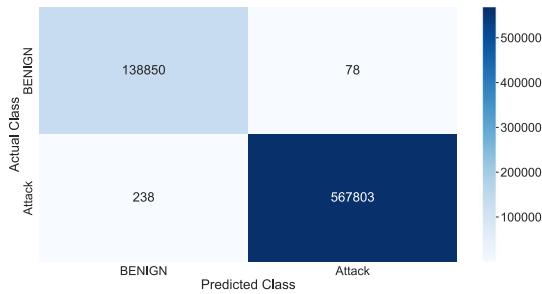
Nevertheless, while these metrics showcase the model's advantageous attributes, a thorough analysis unveils several significant constraints. For instance, there exists a slight diminishment in performance with the MQTT-DDoS-Publish_Flood class, which attained a TPR of 99.76%, an FNR of 0.24%, and an F -score of 99.88%. Similarly, the MQTT-DoS-Connect_Flood class achieved a PPV of 99.70% and an F -score of 99.85%. Furthermore, ARP_Spoofing attained a TPR of 97.60%, FNR of 2.40%,



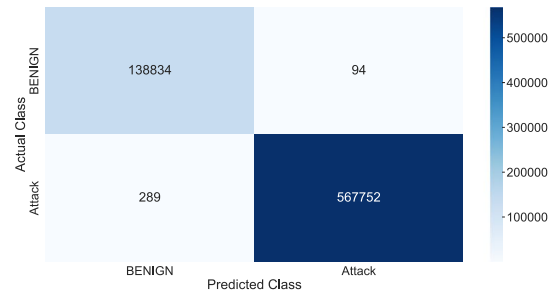
(a) The model accuracy curve illustrating the efficacy of the model in accurately predicting outcomes for both the training and validation data.



(b) Model loss curve representing the error and distance between the true values and the model's predicted values on both the training and validation data.



(c) Binary-class confusion matrix depicting distribution of the predictive performance of DCNN



(d) Binary-class confusion matrix depicting distribution of the predictive performance of DNN

Fig. 6 CICIDS-2017 performance analysis in terms of binary-class classification (2 classes). The analysis includes visualizing accuracy and loss curves, alongside the confusion matrix of the proposed model, to assess the model's learning and improvement over time

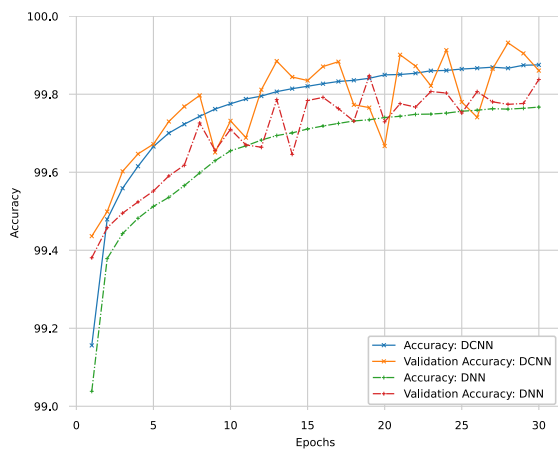
PPV of 93.08%, and F -score of 95.25%, while the Benign class recorded a TPR of 99.44%, FNR of 0.56%, PPV of 99.81%, and an F -score of 99.63%. Additionally, the MQTT-Malformed_Data class obtained a TPR of 91.34%, FNR of 8.66%, PPV of 96.47%, and an F -score of 93.83%. Conversely, one can discern a significant decline in performance with the Recon-OS-Scan class achieving a PPV of 74.08% and an F -score of 84.44%, while the Recon-Ping_Sweep class recorded a TPR of 80.89%, an FNR of 19.11%, PPV of 88.35%, and an F -score of 84.45%. In a similar vein, the Recon-Port_Scan class attained a TPR of 93.94%, an FNR of 6.06%, PPV of 97.81%, and an F -score of 95.84%, whereas the Recon-VulScan class exhibited the poorest performance with a TPR of 37.20%, an FNR of 62.80%, and an F -score of 54.04%. This variation can be attributed to the relatively limited number of instances within the testing dataset that pertain to this specific attack type.

Nonetheless, it is essential to emphasize the resilience of the model's overall performance. To facilitate a more exhaustive comprehension of the model's efficacy, the corresponding confusion matrix is illustrated in Fig. 7c and d. Furthermore, Fig. 7a and b represent the model's performance with respect to both accuracy and loss metrics.

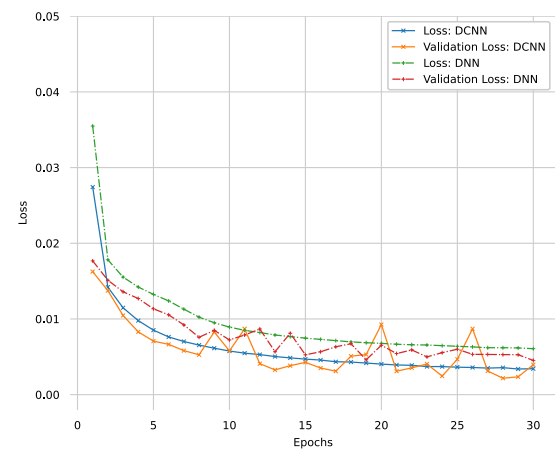
However, in the context of binary-class classification, one can observe a significant enhancement in the overall performance metrics associated with the model's predictive capabilities as outlined in Table 9. It is particularly noteworthy that both of the distinct classes within the binary classification attained remarkably high values for TPR, TNR, PPV, F -score, and accuracy, all exceeding the remarkable threshold of 99%, while simultaneously maintaining very low rates for both FPR and FNR. Also, when binary-class classification is applied to the CICIoMT2024 dataset, the proposed model exhibits a high level of performance

Table 8 Multi-class classification performance analysis with CICIoMT2024

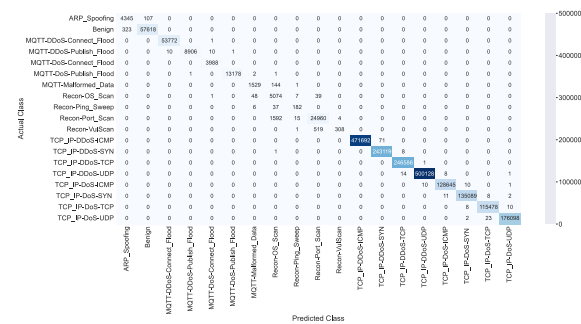
Class	Model	TPR	TNR	FPR	FNR	PPV	F1 measure	Accuracy
ARP_Spoofing	DNN DCNN	86.59% 97.60%	99.98% 99.99%	0.02% 0.01%	13.41% 2.40%	90.68% 93.08%	88.59% 95.29%	99.95% 99.98%
Benign	DNN DCNN	99.29% 99.44%	99.97% 99.99%	0.03% 0.01%	0.71% 0.56%	98.93% 99.81%	99.11% 99.63%	99.95% 99.98%
MQTT-DDoS-Connect_Flood	DNN DCNN	99.97% 100.00%	100.00% 100.00%	0.00% 0.00%	0.03% 0.00%	99.82% 99.98%	99.89% 99.99%	99.99% 100.00%
MQTT-DDoS-Publish_Flood	DNN DCNN	98.95% 99.76%	100.00% 100.00%	0.00% 0.00%	1.05% 0.24%	99.79% 99.99%	99.36% 99.88%	99.99% 100.00%
MQTT-DDoS-Connect_Flood	DNN DCNN	97.64% 100.00%	100.00% 100.00%	0.00% 0.00%	2.36% 0.00%	99.49% 99.70%	98.56% 99.85%	99.99% 100.00%
MQTT-DDoS-Publish_Flood	DNN DCNN	99.86% 99.97%	100.00% 100.00%	0.00% 0.00%	0.14% 0.03%	99.37% 99.99%	99.61% 99.98%	100.00%
MQTT-Malformed_Data	DNN DCNN	96.24% 91.34%	99.99% 100.00%	0.01% 0.00%	3.76% 8.66%	87.84% 96.47%	91.85% 93.83%	99.99% 99.99%
Recon-OS_Scan	DNN DCNN	83.46% 98.16%	99.97% 99.92%	0.03% 0.08%	16.54% 1.84%	87.12% 74.08%	85.25% 84.44%	99.93% 99.91%
Recon-Ping_Sweep	DNN DCNN	84.00% 80.89%	100.00% 100.00%	0.00% 0.00%	16.00% 19.11%	78.75% 88.35%	81.29% 84.45%	100.00%
Recon-Port_Scan	DNN DCNN	96.95% 93.94%	99.95% 99.97%	0.05% 0.03%	3.05% 6.06%	96.08% 97.81%	96.51% 95.84%	99.92% 99.90%
Recon-VulScan	DNN DCNN	50.60% 37.20%	99.99% 100.00%	0.01% 0.00%	49.40% 62.80%	67.26% 98.72%	57.75% 54.04%	99.97% 99.98%
TCP_IP-DDoS-ICMP	DNN DCNN	100.00% 99.98%	100.00% 100.00%	0.00% 0.00%	0.00% 0.02%	100.00% 100.00%	100.00% 99.99%	100.00%
TCP_IP-DDoS-SYN	DNN DCNN	99.99% 100.00%	100.00% 100.00%	0.00% 0.00%	0.01% 0.00%	100.00% 99.97%	99.99% 99.98%	100.00%
TCP_IP-DDoS-TCP	DNN DCNN	99.99% 100.00%	100.00% 100.00%	0.00% 0.00%	0.01% 0.00%	100.00% 99.99%	99.99% 100.00%	100.00%
TCP_IP-DDoS-UDP	DNN DCNN	100.00% 100.00%	100.00% 100.00%	0.00% 0.00%	0.00% 0.00%	99.99% 100.00%	100.00% 100.00%	100.00%
TCP_IP-DoS-ICMP	DNN DCNN	99.99% 99.98%	100.00% 100.00%	0.00% 0.00%	0.01% 0.02%	100.00% 99.99%	100.00% 99.98%	100.00%
TCP_IP-DoS-SYN	DNN DCNN	99.96% 99.98%	100.00% 100.00%	0.00% 0.00%	0.04% 0.02%	100.00% 99.99%	99.98% 99.98%	100.00%
TCP_IP-DoS-TCP	DNN DCNN	99.99% 99.98%	100.00% 100.00%	0.00% 0.00%	0.01% 0.02%	99.95% 99.97%	99.97% 99.98%	100.00%
TCP_IP-DoS-UDP	DNN DCNN	99.99% 99.99%	100.00% 100.00%	0.00% 0.00%	0.01% 0.01%	99.98% 99.99%	99.99% 99.99%	100.00%



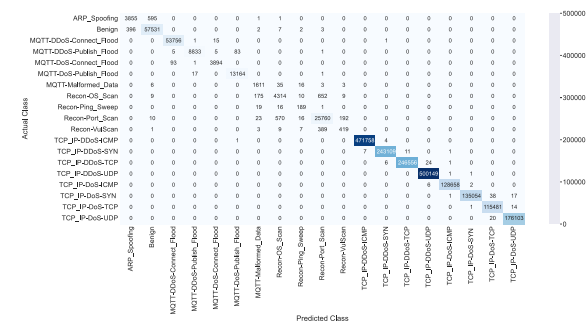
(a) The model accuracy curve illustrating the efficacy of the model in accurately predicting outcomes for both the training and validation data.



(b) Model loss curve representing the error and distance between the true values and the model's predicted values on both the training and validation data.



(c) Multi-class confusion matrix depicting distribution of the predictive performance of the DCNN



(d) Multi-class confusion matrix depicting distribution of the predictive performance of the DNN

Fig. 7 CICIoMT2024 performance analysis in terms of multi-class classification (19 classes). The analysis includes visualizing accuracy and loss curves, alongside the confusion matrix of the proposed model, to assess the model's learning and improvement over time

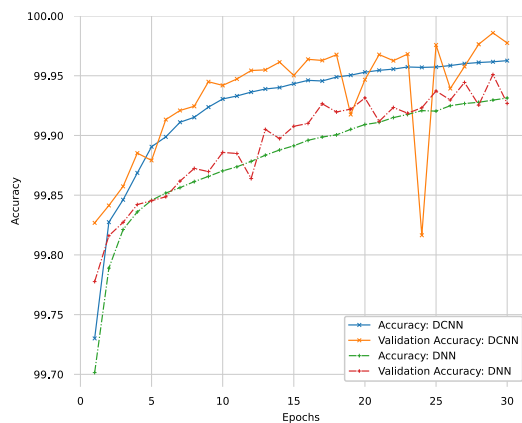
Table 9 Binary-class classification performance analysis with CICIoMT2024

Class	Model	TPR	TNR	FPR	FNR	PPV	F1 measure	Accuracy
Attack	DNN DCNN	99.93% 100.00%	99.69% 99.32%	0.31% 0.68%	0.07% 0.00%	99.99% 99.98%	99.96% 99.99%	99.93% 99.98%
Benign	DNN DCNN	99.69% 99.32%	99.93% 100.00%	0.07% 0.00%	0.31% 0.68%	97.60% 99.82%	98.63% 99.57%	99.93% 99.98%

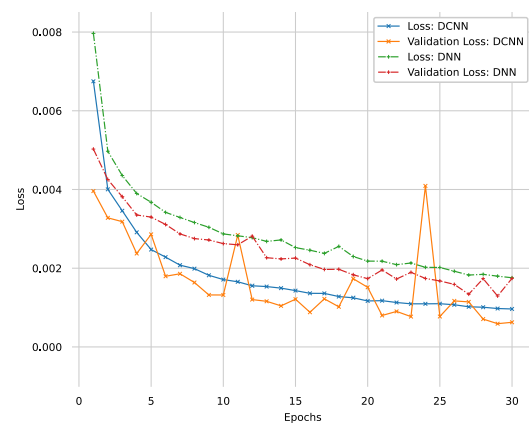
in this context, further validating their effectiveness. The corresponding confusion matrix can be found in Fig. 8c and d, while Fig. 8a and b provide insights into the model's performance in terms of both model accuracy and model loss (Table 10).

As a final assessment of the outcomes derived from the process of evaluation, Table 11 provides a comprehensive analysis of the overall performance of the model

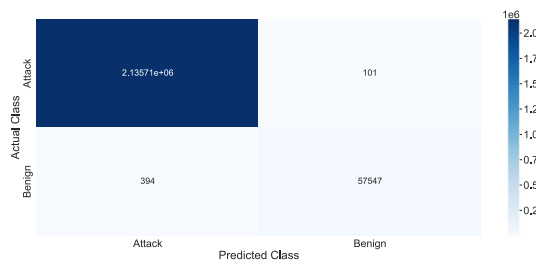
proposed by our work as compared to the state-of-the-art models on the CICIoT2023 dataset. Notwithstanding, the obtained results indicate that our model, when compared to the related-work on the CICIoT2023 dataset, achieves a level of accuracy and *F*-score and PPV that surpasses that of [33], although, it falls slightly short in terms of accuracy and TPR. Also, it outperforms that of [31] in terms of accuracy, PPV, and *F*-score, while it exhibits a



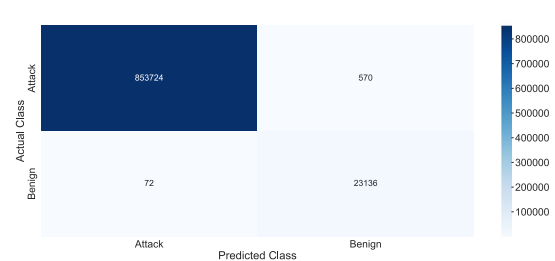
(a) The model accuracy curve illustrating the efficacy of the model in accurately predicting outcomes for both the training and validation data.



(b) Model loss curve representing the error and distance between the true values and the model's predicted values on both the training and validation data.



(c) Binary-class confusion matrix depicting distribution of the predictive performance of DCNN



(d) Binary-class confusion matrix depicting distribution of the predictive performance of DNN

Fig. 8 CICIoMT2024 performance analysis in terms of binary-class classification (2 classes). The analysis includes visualizing accuracy and loss curves, alongside the confusion matrix of the proposed model, to assess the model's learning and improvement over time

Table 10 Performance benchmarking of the proposed model versus other related work with CICIoMT2024 dataset

Ref	Model	Dataset	PPV	TPR	F1 measure	Accuracy
[34]	Logistic regression	CICIoMT2024	B: 95.20% 95.90%	B: 94.00% 96.10%	B: 94.60% 95.90%	B: 99.50% 99.60%
	Adaboost DNN		95.60% 97.10%	94.80% 95.10%	95.20% 96.10%	99.60% 99.60%
	Random Forest		M: 54.70% 14.40%	M: 47.10% 23.80%	M: 43.20% 14.10%	M: 72.70% 42.20%
			64.90% 69.10%	55.30% 57.70%	52.20% 55.10%	72.90% 73.30%
Proposed	DNN DCNN	CICIoMT2024	B: 99.93% 99.98%	B: 99.93% 99.98%	B: 99.93% 99.98%	B: 99.93% 99.98%
			M: 99.84% 99.86%	M: 99.84% 99.86%	M: 99.84% 99.86%	M: 99.84% 99.86%

minor deficiency with respect to TPR. However, it surpasses that of [37] in terms of all metrics. Nevertheless, our model still exhibit exceptional performance, particularly in terms of PPV and F-score metrics, surpassing those of the State-of-the-Art.

Table 12 illustrates the comparison between our work and the state-of-the-art models on the CICIDS-2017 dataset. It is important to note that our proposed model surpasses the related work in all aspects, particularly in terms of *F*-score in both binary and multi-class classification. While [26] manages to achieve a high level

Table 11 Performance benchmarking of the proposed model versus other related work with CICIOT2023 dataset

Ref	Model	Dataset	PPV	TPR	F1 measure	Accuracy
[30]	Blending	CICIOT2023 IoTID20	M: 98.51% 100.00%	M: 99.63% 100.00%	M: 99.07% 100.00%	M: 99.51% 100.00%
[31]	IoT-PRIDS	CICIOT2023	M: 93.84%	M: 99.71%	M: 95.29%	M: 98.74%
[33]	Logistic regression	CICIOT2023	B: 86.31% 82.54%	B: 89.04% 79.70%	B: 87.62% 81.05%	B: 98.90% 98.17%
	Perception Adaboost		96.56% 96.53%	94.73% 96.51%	95.62% 96.52%	99.58% 99.68%
	Random Forest DNN		94.75%	93.32%	94.03%	99.44%
			M: 51.24% 52.39%	M: 69.60% 65.91%	M: 53.94% 55.51%	M: 83.16% 86.63%
			46.49% 70.54%	48.77% 91.00%	36.86% 71.92%	35.13% 99.43%
			67.94%	90.66%	69.72%	99.11%
[37]	CKAN	CICIOT2023 NSL_KDD	B: 99.81% 98.36%	B: 99.40% 98.90%	B: 99.60% 98.63%	B: 99.22% 98.71%
		TONIoT	99.95%	99.91%	99.93%	99.93%
			M: 98.84% 99.20%	M: 98.84% 99.20%	M: 98.84% 99.20%	M: 98.84% 99.20%
			93.30%	93.30%	93.30%	93.30%
Proposed	DNN DCNN	CICIOT2023	B: 99.52% 99.50%	B: 99.52% 99.50%	B: 99.52% 99.50%	B: 99.52% 99.50%
			M: 99.45% 99.25%	M: 99.45% 99.25%	M: 99.45% 99.25%	M: 99.45% 99.25%

of performance in terms of accuracy, wherein we have achieved scores above 99.94%.

A comparison between our work and the state-of-the-art models on the CICIOT2024 dataset is outlined in Table 10. It is important to note that our proposed model surpasses the related work in all aspects in both binary and multi-class classification.

In summary, the analysis conducted on both the CICIDS-2017 and CICIOT2023 datasets clearly demonstrates the effectiveness of the proposed model in both multi-class and binary-class classification approaches. The obtained results, supported by the visual representations and statistical evaluations, highlight the remarkable accuracy and robustness of the model. Thus signifying their potential for real-world implementation and contribution to enhanced detection and prevention of diverse attack types in the cybersecurity field.

Table 13 presents time analysis of fitting and inference times for both binary and multi-class classification across the three datasets: CICIOT2024, CICIOT2023, and CICIDS-2017. The results indicate that fitting times vary between multi-class and binary classifications, with the CICIOT2024 dataset showing fitting times of 356 s for multi-class and 390 s for binary-class. Significantly, the CICIDS-2017 dataset displays the minimal fitting times, recorded at 147 s for multi-class and 138 s for binary-class, whereas the CICIOT2023 dataset shows the highest fitting times of 562 s and 469 s, respectively. Regarding inference times, the CICIOT2024 dataset reflects a multi-class inference time of 127.18 s compared to 119.75s for binary-class. The CICIOT2023 dataset reports an inference times of 238.67 s and 180.88 s, respectively.

Notably, the CICIDS-2017 dataset demonstrates the lowest inference times of 44.60 s and 41.72 s for multi-class and binary-class, respectively.

5 Discussion

Many different machine learning models have been previously developed to address the issue of network intrusion detection from different perspectives. In this paper, our objective is to take advantage of the power of combining CNNs and DNNs in one model in order to improve the overall NIDS system performance and accuracy. When this new hybrid model is trained using three different datasets, it shows a better performance for both binary and multi-class classification.

However, despite these strengths, some limitations emerge, particularly in the detection of certain classes such as Brute-Force and Web-based, that are frequently misclassified as benign, recon, or spoofing, leads to a minor degradation in performance for these classes. For instance, Brute-Force presents a concerning TPR of only 24.94%, which indicates that the model struggles to identify this attack effectively, despite a high TNR of 100%. Such misclassifications highlight potential issues in the training data or suggesting that the model may not be adequately capturing the unique characteristics of these attacks, that may lead to underperformance in identifying Brute-Force attacks. Further considering this aspect in future model development by comparing the model response across different datasets will indeed enhance its accuracy.

Similarly, Web-based exhibits a low TPR of just 12.82% and a high FNR of 87.18%, indicating a gap in the detection capabilities. Such misclassifications also

Table 12 Performance benchmarking of the proposed model versus other related work with CICIDS-2017 dataset

Ref	Model	Dataset	PPV	TPR	F1 measure	Accuracy
[5]	CNN-LSTM	CICIDS-2017	-	M: 97.21%	M: 93.32%	M: 98.67%
[6]	Canet	NSL-KDD UNSW-NB15		B: 99.90% 97.89%	-	B: 99.79% 97.71%
		CICIDS-2017 CICDDoS-2019		99.37% 99.96%		99.67% 99.95%
				M: 99.73% 98.93%		M: 99.77% 89.39%
				99.82% 99.97%		99.88% 99.58%
[7]	OE-IDS	UNSW-NB15	B: 98.00% 99.70%	B: 97.10% 100.00%	B: 70.50% 99.90%	B: 98.60% 99.90%
	OE-IDS+SMOTE	CICIDS2017	97.30% 97.50%	96.00% 99.40%	96.70% 98.40%	98.00% 98.40%
[22]	DNN	NSL-KDD UNSW-NB15	B: 99.94% 95.00%	B: 98.81% 98.95%	B: 99.37% 96.93%	B: 99.84% 89.03%
		CICIDS 2017	99.85%	99.94%	99.89%	99.80%
[23]	Two-phase IDS	NSL-KDD UNSW-NB15	B: 96.00% 88.10%	B: 99.10% 88.10%	B: 97.50% 88.10%	B: 97.10% 86.90%
		CICIDS-2017	95.40%	97.51%	96.44%	98.59%
[24]	CNN	CICIDS-2017	-	M: 99.29%	-	M: 99.64%
[25]	RTIDS	CICIDS-2017 CIC-DDoS2019	M: 98.98% 98.82%	M: 98.83% 98.66%	M: 99.17% 98.48%	M: 99.35% 98.58%
[26]	LSTM-AE	CICIDS-2017 CICIDS-2018	M: 99.99% 99.07%	M: 99.99% 99.10%	M: 99.99% 99.02%	M: 99.99% 99.10%
[27]	PyDSC-IDS	NSL-KDD CICIDS-2017	M: 83.55% 90.73%	M: 81.63% 97.81%	M: 80.63% 94.13%	M: 81.63% 97.60%
[28]	PDAE	KDDCup99 CICIDS2017	-	-	-	B: 99.93% 99.37%
		UNSW-NB15				100.00%
						M: 99.92% 99.43%
						99.84%
[29]	LightGBM + RNDAE	NSL-KDD CICIDS-2017	-	M: 95.90% 97.20%	M: 95.10% 96.40%	M: 96.50% 97.80%
		CICIDS-2018		98.20%	97.40%	98.80%
[32]	CNN-LSTM	CICIDS 2017 UNSW-NB15	B: 99.56% 94.69%	B: 99.70% 94.53%	B: 99.60% 94.77%	B: 99.64% 93.78%
		WSN-DS	98.18%	98.14%	98.00%	99.67%
			M: 99.84% 81.59%	M: 99.95% 82.41%	M: 99.98% 80.87%	M: 99.60% 81.83%
			98.72%	98.83%	98.44%	98.35%
[35]	DT and CatBoost	CICIDS-2017	B: 99.82% M: 99.36%	B: 99.82% M: 99.42%	B: 99.82% M: 99.32%	B: 99.82% M: 99.42%
[36]	DIS-IoT	ToN-IoT CICIDS-2017	B: 99.40% 95.90%	B: 99.40% 97.60%	B: 99.40% 96.70%	B: 99.60% 98.70%
		SWaT	99.70%	99.90%	99.80%	99.60%
			M: 99.70% 98.70%	M: 99.70% 98.70%	M: 99.70% 98.60%	M: 99.70% 98.70%
Proposed	DNN DCNN	CICIDS-2017	B: 99.95% 99.96%	B: 99.95% 99.96%	B: 99.95% 99.96%	B: 99.95% 99.96%
			M: 99.94% 99.96%	M: 99.94% 99.96%	M: 99.94% 99.96%	M: 99.94% 99.96%

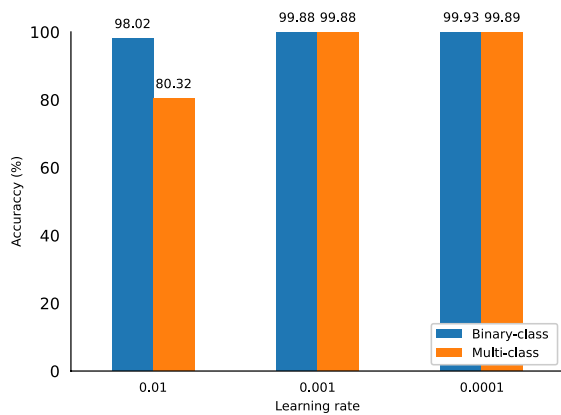
Table 13 Runtime analysis of the proposed model for training as well as inference stages

Dataset	Case study	Fitting time (per epoch)	Inference time (25%)	Inference time (per record)
CICIoMT2024	Multi-class	356 s	127.18 s	0.071042024 ms
	Binary-class	390 s	119.75 s	0.066891668 ms
CICIoT2023	Multi-class	562 s	238.67 s	0.080925604 ms
	Binary-class	469 s	180.88 s	0.061330805 ms
CICIDS-2017	Multi-class	147 s	44.60 s	0.063086217 ms
	Binary-class	138 s	41.72 s	0.059012489 ms

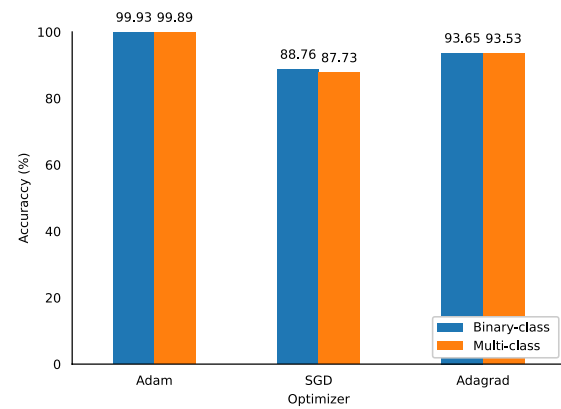
Experiments conducted using the configuration listed in Table 3

highlight potential issues in the training data or feature selection that may lead to inefficiency in identifying these specific attacks. In addition, The results for Recon and Spoofing attacks also reveal challenges, with TPR of 65.34% and 74.67%, respectively, suggesting that

while the model performs well overall, certain attack types are not being effectively recognized. Nonetheless, it is noteworthy that despite the model making these errors due to the similarities in data patterns, the



(a) The model accuracy scores illustrating the effect of variations in learning rate on the model performance in the context of binary and multi-class classification.



(b) The model accuracy scores illustrating the effect of variations in optimizers on the model performance in the context of binary and multi-class classification.

Fig. 9 Comparison of the model accuracy when two hyper-parameters are varied: (a) learning rate and (b) optimizers. The value of learning rate of 0.0001 provides the best accuracy for both binary and multi-class classification, while Adam optimizer algorithm offers the best accuracy compared to other considered algorithms

classification process is successful in the majority of other cases.

Furthermore, when dealing with class imbalance, there are many techniques that can be employed to address the issue, including oversampling the minority class, under-sampling the majority class, and using class weights. However, in this study, we deliberately chose to focus on data standardization to reduce the effect of class imbalance. By standardizing the data, we ensured that all features have equal importance, which helped to mitigate the impact of class imbalance. Additionally, we employed alternative metrics, such as *F1*-score, precision, and recall to evaluate the model's performance for each class. These metrics provide a more comprehensive understanding of the model's ability to detect minority class instances.

When training a neural network, key hyperparameters play a crucial role in shaping how the network learns and performs. In our model, we experimented with different learning rates and optimizers to identify the optimal settings that yield the best accuracy and parameter optimization. Figure 9 illustrates how model accuracy responds to various learning rates and optimizer choices, while Figs. 10 and 11 provide detailed views of these effects as training progresses across epochs.

We experimented with three learning rates: 0.01, 0.001, and 0.0001. As shown in Fig. 9a, a learning rate of 0.0001 achieved the highest accuracy among these

values. Figure 10a through d further confirm that 0.0001 optimizes both accuracy and learning loss. This learning rate is also commonly used in similar models across the literature. Lower values than 0.0001 were found to negatively impact accuracy and loss. Thus, we selected 0.0001 as the optimal learning rate for training our model in this study.

Similarly, various optimization algorithms are commonly used to train deep neural networks effectively. In this study, we experimented with three optimizers: Adam, SGD, and Adagrad to identify the best fit for our model. As shown in Fig. 9b, the Adam optimizer achieved the highest accuracy among the options tested. Figure 11a through d further confirm Adam's consistent performance improvement with each training epoch.

Finally, while we did not have the opportunity to test the model in a real network environment (see Sect. 6), the runtime measurements presented in Table 13 indicate that the model could operate within a real network without introducing significant latency. This potential will be further validated in future studies.

6 Conclusion and future work

In this paper, we proposed an NIDS based on deep learning, utilizing a hybrid deep convolutional neural network model. This model combines two deep learning paradigms, namely convolutional neural network and deep

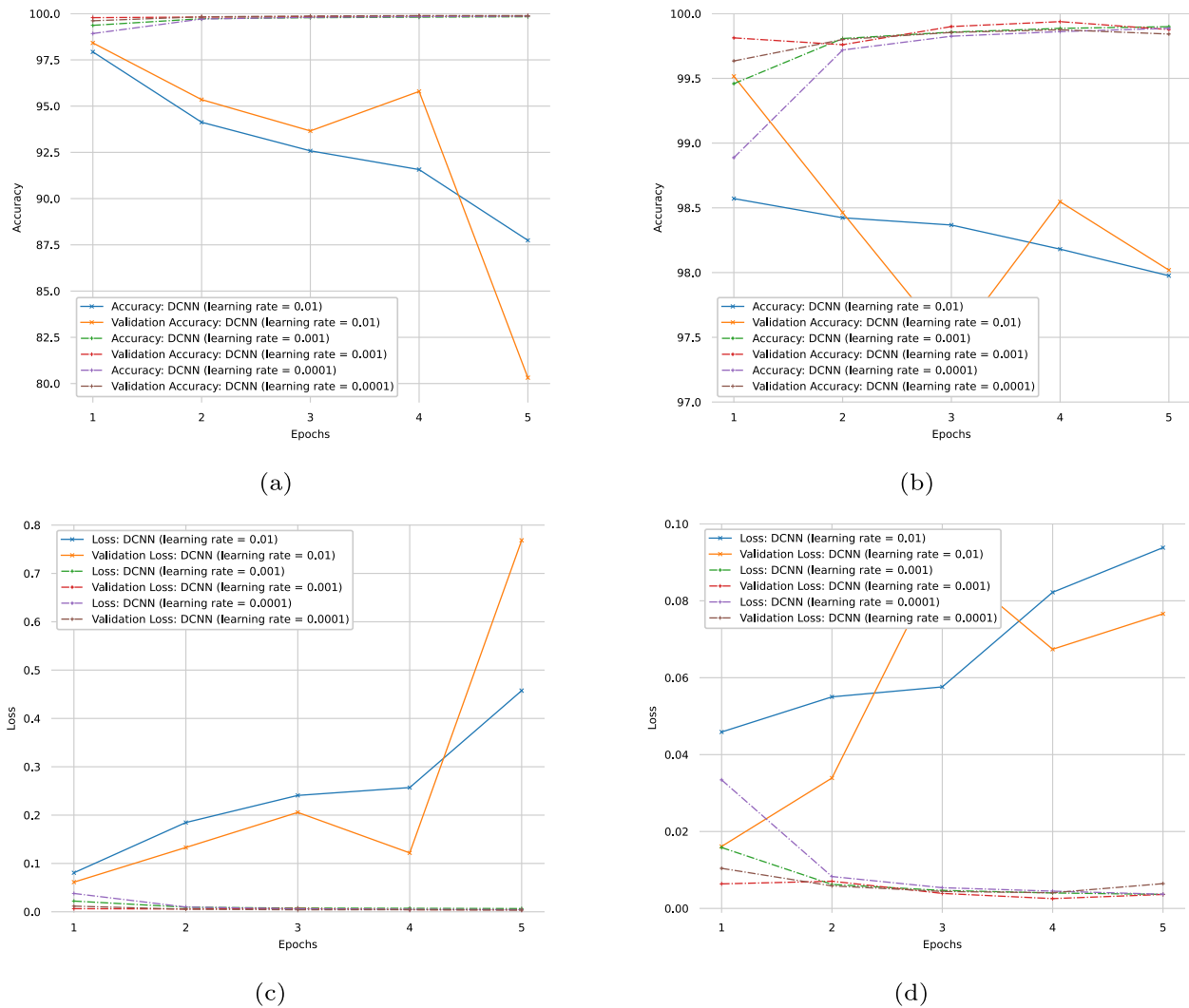


Fig. 10 Comparison of the model accuracy and loss (y-axis) with different learning rate values and different number of epochs (x-axis): **(a)** epochs versus accuracy for multi-class classification, **(b)** epochs versus accuracy for binary-class classification, **(c)** epochs versus loss for multi-class classification, and **(d)** epochs versus loss for binary-class classification

neural network. To evaluate the performance of the proposed model, experiments were conducted using three benchmark datasets. In order to thoroughly analyze and evaluate the model performance, two distinct scenarios were taken into consideration, namely multi-class classification as well as binary-class classification.

The findings indicate that the model attained remarkably high F -score, accuracy, positive predictive value, and true positive rate. The scores obtained for binary and multi-class classifications is ranging from 99.25 to 99.98% depending on the dataset used, which further improved existing models in the literature. By leveraging the power of deep learning, our model is able to effectively analyze complex patterns in network traffic data and accurately identify potential intrusions. This will contribute to the

field of network security and has the potential to greatly improve the overall security of interconnected networks and systems. The model implementation can be found at this link <https://github.com/AhmedShebl13/NIDS>.

The presented model offers several avenues for future development. First, the detection of minor classes, as elaborated in the Sect. 5, could be enhanced to improve the classification accuracy. Second, evaluating the model with additional benchmark datasets would strengthen its robustness. Third, integrating innovative solutions, such as blockchain, could further expand the model's applications. Fourth, to provide insight into the model's performance in a real network environment, we have relied on its detection runtime. However, testing the model by deploying it in an actual environment

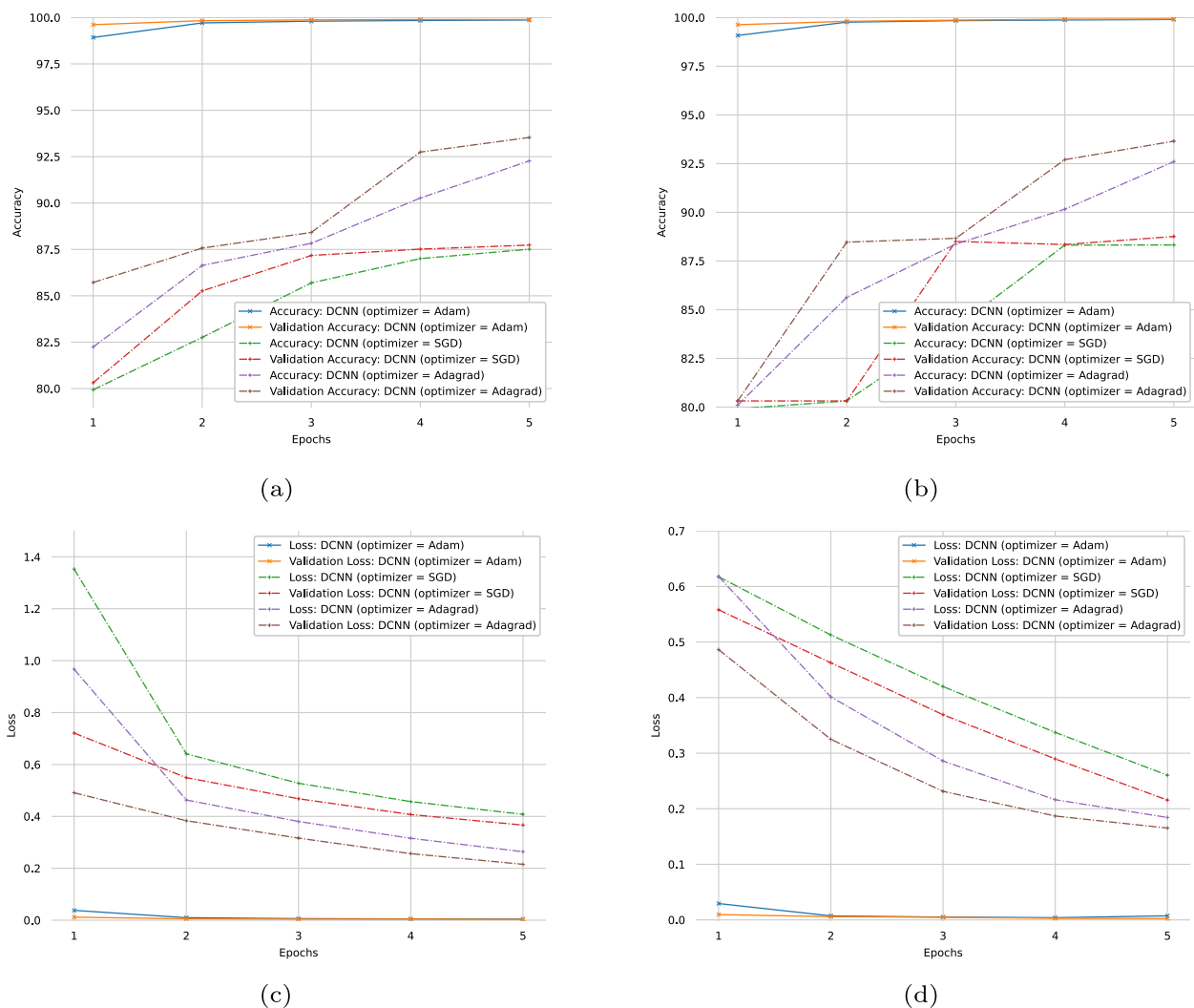


Fig. 11 Comparison of the model accuracy and loss (y-axis) with different optimizers and different number of epochs (x-axis): **(a)** epochs versus accuracy for multi-class classification, **(b)** epochs versus accuracy for binary-class classification, **(c)** epochs versus loss for multi-class classification, and **(d)** epochs versus loss for binary-class classification

would give a more accurate assessment of its effectiveness and reveal additional areas for improvement. Lastly, like many deep learning models, the current model operates as a “black box,” lacking transparency in how specific classifications are determined. Investigating methods to explain the model’s outputs would be a valuable direction for future research.

Abbreviations

IoT	Internet of Things
NIDS	Network intrusion detection system
AI	Artificial intelligence
ML	Machine learning
DL	Deep learning
CNN	Convolutional Neural Network
DNN	Deep neural network
LSTM	Long short-term memory
PSO	Particle swarm optimization

GA	Genetic algorithm
DCNN	Deep convolutional neural network
CIC	Canadian Cybersecurity Institute
ANN	Artificial neural network
ReLU	Rectified linear unit
FCL	Fully connected layers
DR	Detection rate
TPR	True positive rate
PPV	Positive predictive value
TNR	True negative rate

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Authors’ contributions

Ahmed Shebl proposed the idea, Mostafa Herajy refined the idea. E. Elsedimy, A Ismail, and A Salama validate the model. Ahmed Shebl wrote the draft. Mostafa Herajy and E. Elsedimy revised and improved the manuscript. All the authors read and improved the manuscript.

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Data availability

No datasets were generated or analyzed during the current study.

Declarations**Ethical approval and consent to participate**

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare that they have no competing interests.

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